

INTRODUCTION TO NETWORKING

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LESSON REWARD

1. Understanding what a network is & its components
2. Be able to understand types of topologies
3. Be able to understand types of transmission medium
4. Be able to create your first network using Cisco Packet tracer
5. Be able to configure a wireless router

REQUIREMENT

1. Laptop (Recommended)
 - with at least 4GB RAM & 2GB ROM space or above
2. Cisco Packet Tracer
 - Download at [computernetworknotes](http://computernetworknotes.com)
3. Researching
 - Don't limit yourself, explore.



CISCO PACKET TRACER

Try simulation mode to see how packets travel through your networks.



WHAT IS NETWORKING?

- Networking is simply the practice of connecting two or more computers with a purpose of enabling communication and sharing resources with each other.

Key components

1. Devices: These can be router, switch, computer, phone, server etc
2. Transmission media: This can be a cable (wired) or radio waves (wireless)
3. Protocols: These govern how communication must be done e.g http
4. IP address: Unique identifiers for each device on a network
5. Router: Directs traffic between networks
6. Switch: Connecting devices within the same network

WHAT IS A TOPOLOGY

Topology refers to the arrangement or layout of devices are connected and how data flows between them.

Two dimensions

1. **Physical Topology:** This is simply the actual physical layout of cables, devices and connections.
2. **Logical Topology:** This mimics how devices can be arranged, used for testing connection and feasibilities of network setup.

COMMON TOPOLOGY TYPES

1. Bus Topology

All devices share a single central cable for communication and resource sharing.

2. Ring Topology

Devices are connected in a closed loop, data travels in one direction around the ring.

3. Star

All devices connect to a central hub or switch.
Most common in modern LANs.

WHAT IS A TRANSMISSION MEDIUM

A transmission medium is the physical or logical pathways through which data signals travel from one device to another in a network.

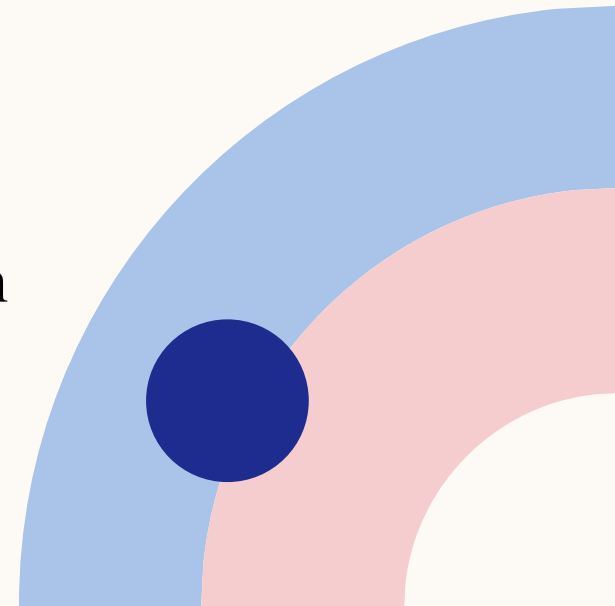
TWO MAJOR CATEGORIES

1. Guided (wired) Media

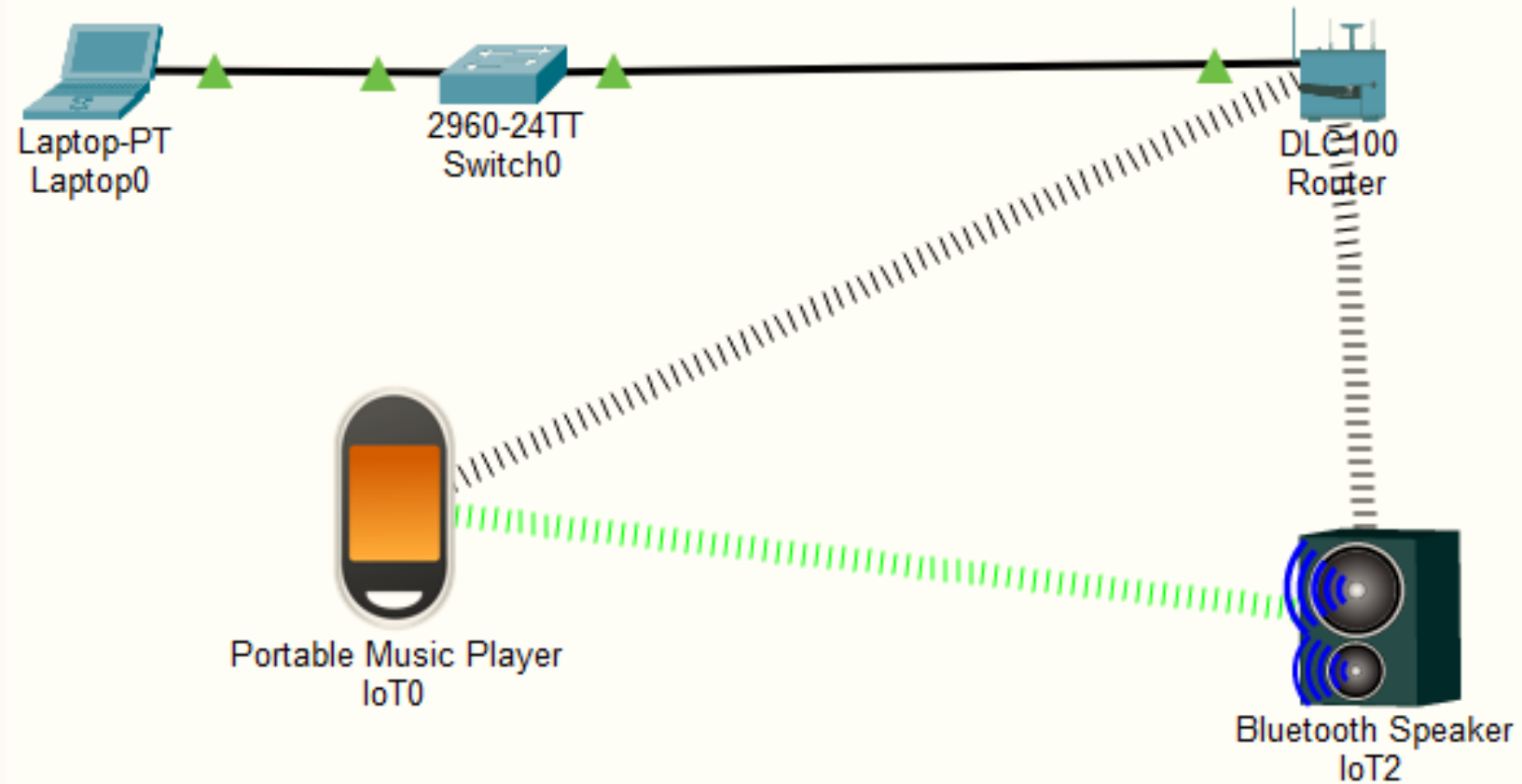
The signal is physically guided along a solid medium e.g coaxial cable, fibre optic cable, ethernet cable etc.

2. Unguided (wireless) Media

The signal travels through free space (air) without a solid medium e.g. radio waves, microwaves, infrared etc



STAR TOPOLOGY





THANK YOU